



ROBOT CHALLENGE

Girls in STEAM, XRP Edition

YOUR MISSION

Program the robot to drive through the maze on its own. Snap blocks together to tell it what to do, in order.



RULES

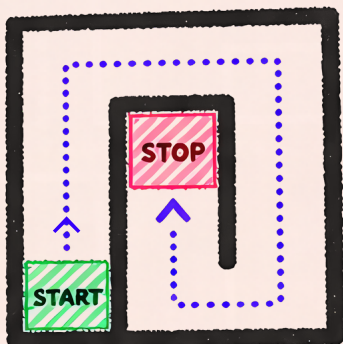
STOP

- 1 Start in the green square.
- 2 Stay between the lines. Only 1 wheel may cross a boundary line at a time.
- 3 Finish in the red square.

HINTS

- 1 Robots drift a bit. Tweak your numbers to fix it.
- 2 Take it one command at a time. Once it works, move on.
- 3 Ask for help!

THE COURSE



INSTRUCTIONS

- 1 **Plan it.** Use the blocks in the Command Key to write down the commands your robot will follow. Measure each distance in inches with a tape measure.
- 2 **Build it.** On the laptop, drag the blocks onto the dotted workspace to match your plan. Blocks run from top to bottom.
- 3 **Place it.** Put the robot in the green start square, pointing the way your plan starts.
- 4 **Run it.** Let go of the robot, then click RUN. Keep your hands off until it stops.
- 5 **Fix it.** Watch what the robot does, then go back and change your blocks.

COMMAND KEY

- ↑ Drive forward ____ inches
- ↓ Drive backward ____ inches
- ↶ Turn left ____ degrees
- ↷ Turn right ____ degrees
- ⬛ Stop motors

Left and right are the robot's left and right.